Solutions: Day 36 Practice Problems (Hinglish)

Day 36 Deep Learning challenges ke step-by-step mathematical calculations, PyTorch code, aur complete solutions.

Solution 1: Manual Forward Pass & PyTorch Verification

1. Manual Linear Sum (z)

Weighted sum ka formula:

$$z = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b$$

Values substitute karein:

$$z = (0.40 \times 0.60) + (0.70 \times 0.75) + (0.50 \times 0.80) + (-0.60 \times 0.50) + (-0.15)$$

$$z = 0.240 + 0.525 + 0.400 - 0.300 - 0.150$$

$$z = 0.7150$$

2. Sigmoid Activation ($\hat{y}$)

$$\sigma(z) = \frac{1}{1 + e^{-0.7150}} = \frac{1}{1 + 0.48919} \approx \mathbf{0.6715}$$

- **Faisla**: Kyunki probability $0.6715 \geq 0.50$ hai, isliye patient **High Risk (Class 1)** classify hoga (67.15% confidence).

3. PyTorch Verification Code

```
import torch
import torch.nn as nn

# Inputs tensor
inputs = torch.tensor([0.60, 0.75, 0.80, 0.50])

# Neuron banayein (4 inputs, 1 output)
neuron = nn.Linear(in_features=4, out_features=1)

# Manual weights assign karein
with torch.no_grad():
    neuron.weight.copy_(torch.tensor([[0.40, 0.70, 0.50, -0.60]]))
    neuron.bias.copy_(torch.tensor([-0.15]))

# Forward pass
z = neuron(inputs)
y_hat = torch.sigmoid(z)

print("Linear sum z :", z.item())
print("Sigmoid y_hat:", y_hat.item())
```

Output: `Linear sum z : 0.7150` `Sigmoid y_hat: 0.6715` (Manual calculation aur PyTorch 100% exact match hain!)

Solution 2: Non-Linearity aur XOR Problem

1. Mathematical Contradiction ka Proof

Charon XOR points ke liye linear equations:

1. $(0,0) \to 0 \implies b < 0$
2. $(0,1) \to 1 \implies w_2 + b \geq 0$
3. $(1,0) \to 1 \implies w_1 + b \geq 0$
4. $(1,1) \to 0 \implies w_1 + w_2 + b < 0$

Eq 2 aur Eq 3 ko add karein:

$$(w_1 + w_2 + b) + b \geq 0$$

Lekin Eq 4 kehta hai ki $(w_1 + w_2 + b) < 0$ (negative), aur Eq 1 kehta hai ki $b < 0$ (negative). Do negative numbers ka sum hamesha negative hota hai (< 0), kabhi bhi ≥ 0 nahi ho sakta! Yeh **Contradiction** prove karta hai ki ek single linear neuron XOR kabhi solve nahi kar sakta.

2. Multi-Layer Solution (2 Layers)

XOR ko hum OR aur NAND gate mein tod sakte hain:

$$\text{XOR} = \text{OR}(x_1, x_2) \wedge \text{NAND}(x_1, x_2)$$

- Hidden Neuron 1 (OR gate banata hai): $w_1 = 1, w_2 = 1, b = -0.5$
- Hidden Neuron 2 (NAND gate banata hai): $w_1 = -1, w_2 = -1, b = 1.5$
- Output Neuron (AND gate banata hai): $v_1 = 1, v_2 = 1, b = -1.5$

Yeh 2-layer network XOR ko 100% sahi classify karta hai!

Solution 3: Industrial Architecture Sizing

1. Traditional ML kyun fail hoti hai?

1. **Dimensionality Explosion**: $1024 \times 1024 \times 3$ image me 31 lakh se zyada features hote hain. SVM itne bade dimension me fail ho jata hai.
2. **Handcrafted Features ki Limitation**: Insaan dwara banaye gaye filters (edges, SIFT) rotation aur lighting change hone par fail ho jate hain.
3. **Automated Feature Learning**: Deep Learning raw pixels se khud abstract features seekhti hai.

2. Framework Selection

R&D ke liye **PyTorch** best hai kyunki isme custom layers banana aur debug karna aasan hai. Deployment ke liye PyTorch model ko ONNX ya TensorRT me convert karke NVIDIA Jetson chip par fast chalaya ja sakta hai.

3. VRAM Memory

Batch 32 images ka raw size:

$$32 \times 3 \times 1024 \times 1024 \times 4 \text{ bytes} \approx 402.65 \text{ MB}$$

Training ke waqt intermediate activations aur gradients store karne padte hain, aur Adam optimizer har parameter ka moving average rakhta hai, isliye total memory raw data se 3x-4x zyada (~1.5GB to 2GB) lagti hai.